

WILLIAM FARLESSYOST

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EDUCATION

Purdue University Agricultural and Biological Engineering, Ph.D. <i>Majors: Computational Science & Engineering, Ecological Sciences & Engineering</i>	2025
University of North Carolina Asheville Mechatronics Engineering, B.S.E. (Jointly Awarded from NCSU-UNCA) Applied Mathematics B.A.	2020 <i>Summa cum laude</i> <i>Summa cum laude</i>

WORK EXPERIENCE & RESEARCH

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| Graduate Research Assistant, Purdue University
<i>Sustainable Industrial-Natural Coupled Systems Group, PI: Shweta Singh, Ph.D.</i> | Aug. 2020 - Present |
| <ul style="list-style-type: none">· Applied SINDy for system identification and improving mechanistic models in chemical processes, including an algae batch reactor case study.· Modeled material flow networks using liquid time constant neural networks, performing robustness and sensitivity analysis over climate scenarios, and minimized sensor requirements.· Compared material economies of Indiana and Illinois using PIOT-Hub for rapid mapping of physical input-output tables. | |
| Undergraduate Research Assistant, UNCA.
<i>Undergraduate Research in Controls</i> | Aug. 2019 - Jan. 2020 |
| <ul style="list-style-type: none">· Validated and implemented a quadrotor UAV control system enabling it to track and softly land on a moving base using fiducial-tag recognition and sensor fusion in ROS and Simulink. | |
| Undergraduate Research Assistant, UNCA
<i>Undergraduate Research in Robotics (NSF-Funded)</i> | Aug. 2019 - May 2020 |
| <ul style="list-style-type: none">· Developed an apprenticeship learning system for a UR3e robot arm with RG2 gripper to sort unobserved objects, integrating ROS packages, motion planning with MoveIt!, and simulation in Gazebo. | |
| Mechatronics Senior Design Project, UNCA
<i>NASA Lunabotics Challenge</i> | Aug. 2019 - May 2020 |
| <ul style="list-style-type: none">· Designed and built two autonomous double Ackermann steered mining robots for the 2020 NASA Lunabotics Competition (event canceled due to COVID-19), implementing software for SLAM, path planning, and control. | |
| Undergraduate Research Assistant, University of Nevada, Reno
<i>NSF-REU on Collaborative Human-Robot Interaction</i> | June 2019 - Aug. 2019 |
| <ul style="list-style-type: none">· Developed multi-task convolutional neural networks in TensorFlow for multi-label classification and regression to recognize social traits in facial images. | |
| Undergraduate Research Assistant, UNCA
<i>Undergraduate Research in Robotics</i> | Sept. 2017 - May 2019 |
| <ul style="list-style-type: none">· Developed a Jenga-playing robot for reinforcement learning research, involving mechanical and electrical design, ROS implementation, motion planning with MoveIt!, and simulation in Gazebo. | |

- Designed and built an autonomous robot with a state machine for navigation to complete tasks in an arena.

PUBLICATIONS & PRESENTATIONS

Farlessyost W, Singh S. “Resilience Dynamics in Coupled Natural-Industrial Systems: A Surrogate Modeling Approach for Assessing Climate Change Impacts on Industrial Ecosystems.” *Manuscript in Review*. Journal of Industrial Ecology.

Farlessyost W, Singh S. “Improving Mechanistic Model Accuracy with Machine Learning Informed Physics.” *Poster Presentation, Conference Proceedings*. Foundations of Computer Aided Process Design, 2024.

“Modelling Dynamics of Material Flows in Coupled Industrial Processes Using Data-Driven System Identification.” *Poster Presentation*. Gordon Research Conference on Circular Economy, 2022.

Farlessyost W, Singh S. Reduced order dynamical models for complex dynamics in manufacturing and natural systems using machine learning. *Nonlinear Dynamics*. 2022; 110 (2), 1613-1631

“A Hybrid Mechanistic-Machine Learning Approach to Identify Dynamical Models for Sustainability Assessment of Manufacturing Processes: A Soybean Diesel Process Case Study.” *Poster Presentation*. American Institute of Chemical Engineers Annual Meeting, 2021.

“Sustainability and Resiliency Assessment for Industrial Synergies between Renewable Energy and Chemical Production.” *Oral Presentation*. American Institute of Chemical Engineers Annual Meeting, 2021.

Farlessyost W, Grant K-R, Davis SR, Feil-Seifer D, Hand EM. The Effectiveness of Multi-Label Classification and Multi-Output Regression in Social Trait Recognition. *Sensors*. 2021; 21(12):4127.

“Experimental Validation of a Control Law for an Unmanned Aerial Vehicle Landing on a Moving Base.” *Oral Presentation* (University of North Carolina Asheville Fall Undergraduate Research Symposium, 2019).

“The Effectiveness of Multi-Label Classification and Multi-Output Regression in Social Trait Recognition.” *Invited Presentation* (Mathematical Association of America NC State Dinner, 2019), *Poster Presentation* (University of Nevada, Reno Summer Undergraduate Research Symposium, 2019).

“Vibrational Response of a Wheeled-Legged Robot of Differing Structural and Geometric Attributes.” *Poster Presentation* (University of North Carolina Asheville Spring Undergraduate Research Symposium, 2019).

TEACHING

Graduate Teaching Assistant, Purdue University

Jan. 2021 - May 2021

Statistical Modeling & Quality Enhancement (Purdue ChE-32000)

- Held weekly help sessions, taught course labs, graded HW and exams.

Undergraduate Teaching Assistant, NCSU

Jan. 2020 - May 2020

Principles of Automatic Control (NCSU MAE-435)

- Held two help sessions weekly, facilitated end of semester control design project.

Undergraduate Teaching Assistant, NCSU

Jan. 2019 - May 2019

Intro. to Computer Systems (NCSU ECE-109)

- Taught a weekly 2-hour lab section, held two help sessions weekly.

Undergraduate Teaching Assistant, NCSU
Engineering Thermodynamics (NCSU MAE-201)

Aug. 2018 - Dec. 2018

- Held two help sessions weekly.

Undergraduate Teaching Assistant, NCSU
Analytical Foundations of ECE (NCSU ECE-220)

Jan. 2018 - May 2018

- Taught two 3-hour lab sections every week, held two help sessions weekly, graded weekly lab write-ups and programs.

Undergraduate Teaching Assistant, NCSU
Engineering Statics (NCSU MAE-206)

Jan. 2017 - Dec. 2017

- Held two help sessions weekly, worked as flipped-class facilitator to answer student questions.

TECHNICAL SKILLS & EXPERTISE

Programming Languages:	C/C++, MATLAB, Python, R, Bash/Shell Scripting
Machine Learning Frameworks:	TensorFlow, PyTorch, Scikit-learn, LTC-NN, LSTM, RNN, CNN, SINDy, Fuzzy Learning Methods
Software & Tools:	ASPEN Plus/Dynamics, SimaPro, ROS, Gazebo, MoveIt!, Simulink, Verilog, SolidWorks, BioCro, AutoCAD, Git, SQL
Data Analysis & Modeling:	Statistical Process Control, Multivariate Statistics, Time Series Analysis, Regression Modeling, Traditional System Identification
Domain-Specific Expertise:	DMFA, LCA, Climate Impacts Modeling, Circular Economy Analysis, Process Simulation, Supply Chain Resilience

ACADEMIC ACHIEVEMENTS & AWARDS

Graduate Research Fellowship Program National Science Foundation	2022-2025
Award for Departmental Professionalism and Academic Excellence Joint NCSU-UNCA Mechatronics Department	2020
Top Math Student Award UNCA Department of Mathematics	2020
Meritorious Award SIMIODE Challenge Using Differential Equations Modeling	2019

CLUBS & ORGANIZATIONS

Logistics Committee – Purdue ESE Symposium <i>Member</i>	Aug. 2021 - May 2022
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- Coordination of guest speakers for Purdue's 2022 Ecological Sciences & Engineering (ESE) annual symposium.

Mechatronics Ambassadors Program

April 2017 - May 2020

Chair of University Outreach Committee

- Worked with other departments on campus to provide opportunities and resources to mechatronics students as well as students majoring in other STEM fields.

UNCA IEEE Chapter

Aug. 2016 - May 2020

Member

- Involved with chapter's hardware competition team, skill share events, and engineering outreach on campus.